From Hourly Prices to Quarter-Hour Reconstruction: Experiment History and a Retrieval-Assisted Ensemble

Andrii Zhuravlov, Dobri Trifonov, Michelangelo Bettini, Alessio Barletta

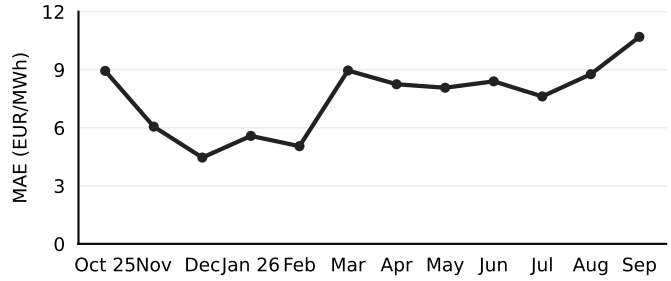


Fig. 1. Monthly flat-baseline MAE across the four zones. The unresolved within-hour variation changes substantially across months.

TABLE I
BASELINES UNDER THE ESTABLISHED PROTOCOLS: MAE

Method	MOSAIC	HORIZON	HARVEST
Flat	7.448	8.228	8.944
MAKIMA	4.453	4.780	5.182
LightGBM	3.806	4.213	4.487
XGBoost	3.811	4.234	4.553
GBM average	3.792	4.200	4.508

EUR/MWh; macro-month means for MOSAIC/HORIZON. Curve-feature GBMs use MOSAIC-selected rounds, frozen for HORIZON/HARVEST; their average has equal weights. Latest-main source: 34edb51. RF metrics were not recovered; dashes are not measured scores.

I. TASK, DATA AND EVALUATION

We reconstructed 96 ordered quarter-hour electricity prices from the complete 24-hour curve of each of four market zones. The supplied hourly mean is a constraint: the model learns the shape inside each hour. We used the official package's 363 complete dates from October 2025 to September 2026, excluding the two DST dates. Training hourly inputs were the means of the four observed quarters. The score averages daily 96-quarter MAE within each zone, then weights the four zones equally. Each serialized hour must satisfy

$$\left| \frac{1}{4} \sum_{q=1}^{4} \widehat{y}_{d,z,h,q} - H_{d,z,h} \right| \leq 10^{-4} \text{ EUR/MWh}.$$

MOSAIC holds out each November–September month in turn; **HORIZON** expands training from November–January and validates on February–September, giving eight folds; **HARVEST** trains on November–September and tests October 2025. We report HORIZON macro-month means. Because the challenge evaluates October 2026, we used October 2025 to assess seasonal transfer. This is an upsampling problem, not forecasting unknown hourly prices: we assume that the quarter-hour profile, conditional on the complete hourly curve, is sufficiently stable across years. This assumption motivates HARVEST but is not verified by one October. October was inspected during development and informed the final choice; it is not an untouched outer test.

II. BASELINES AND INTRA-HOUR STRUCTURE

We decomposed all 139,392 observed prices, after subtracting their hourly means, into orthogonal linear-ramp, curvature and higher-order components. Their shares of squared residual magnitude were **90.45%, 6.18% and 3.37%**. Thus, estimating the intra-hour slope is the main reconstruction task. These are descriptive target-energy shares, not predictive R^2. The target-fitted linear ramp is an oracle diagnostic and cannot be used at inference.

We compared linear, spline and shape-preserving interpolation, including MAKIMA and interpolation of the cumulative hourly-price integral. Our fixed MAKIMA reference samples hour-centered inputs at quarter centers, extrapolates at the boundaries, centers every four-quarter residual and multiplies it by two. Engineered GBM features then captured slopes, curvature and interpolation responses. Their larger gains came from feature geometry rather than hyperparameter tuning. Table I gives the flat, interpolation and tree references under the established protocols; GBM complexity is selected on MOSAIC and frozen for transfer evaluation.

III. TRANSFORMER REPRESENTATIONS AND TARGETS

We first used a bidirectional **96-position masked Transformer**: each hour contributes one known hourly anchor and three mask tokens; the encoder outputs one value at each quarter position. All four outputs are centered around the supplied hour. We then tested a **24-token Transformer** with one known hourly token and four output values per hour. Both process the complete day jointly. Per-day normalization, RoPE and Fourier hour features provided the common backbone; global scales were fitted on training dates only.

For **A0**, let $m_{h,q}$ be the fixed zero-mean MAKIMA residual. The network learns the remaining correction to $y_{h,q} - H_h - m_{h,q}$:

$$\widehat{y}_{h,q} = H_h + m_{h,q} + s\,(z_{h,q} - \bar{z}_h).$$

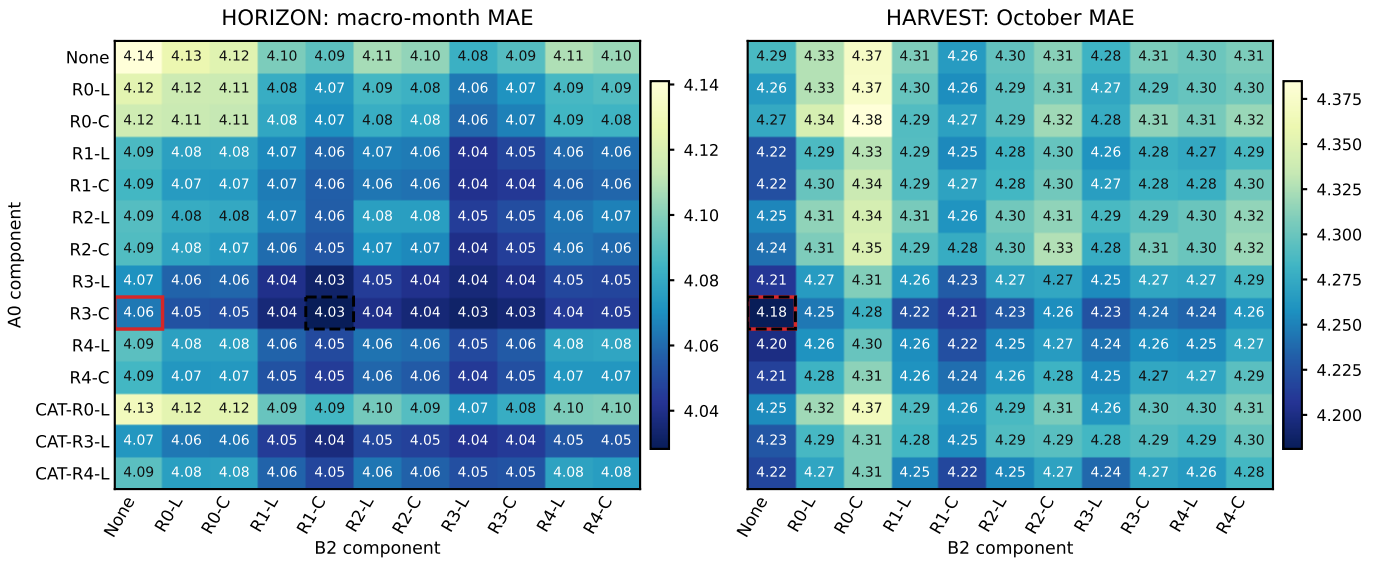


Fig. 2. All 154 fixed 50/50 A0×B2 recipes from the saved retrieval ensemble grid, seed 42. None denotes the original context-free component; L/C denote linear/CNN encoding. Darker cells have lower MAE; each panel has its own scale. Values are rounded to two decimals. Red outlines mark the final R3-CNN + Original recipe; dashed outlines mark the panel minimum. CAT rows use concatenative A0 fusion.

A0 uses four local features: incoming/outgoing differences, curvature and radius-one volatility, plus separate zone/time conditioning. A training-fit gate shrinks the full correction toward the hourly mean.

For **B2**, the target is directly $y_{h,q} - H_h$, without MAKIMA. Its 24-token head predicts upward/downward softmax distributions and a positive amplitude:

$$\widehat{y}_{h,q} = H_h + s\, a_h(p_{h,q} - d_{h,q}), \quad \sum_q p_{h,q} = \sum_q d_{h,q} = 1.$$

Writing g for the training-fitted global scale and s for the floored daily scale, B2's **eleven numerical features** are H_h/s, Δ_h^-/s, Δ_h^+/s, curvature$/s$, Δ_h^-/g, Δ_h^+/g, H_h/g, daily mean$/g$, $\log(s/g)$, and local standard deviations at radii one and two divided by s. Zone and Fourier time conditioning are separate. The 19-channel alternative appends four zone indicators and four time channels. It transferred worse on HARVEST, although it slightly improved replicated HORIZON; eleven features were retained for final B2, not claimed universally superior.

Teacher forcing reveals one true quarter at selected masked positions during training, while evaluation remains fully label-free. It improved the early masked-model comparisons, so we used it in the controlled and replication studies. The final context models disable revelation, and the 24-token models never use it. Alternative losses, flux/potential heads and probabilistic constructions did not displace ordinary MAE training with direct constrained outputs.

IV. Replication Led to Complementary Ensembles

Increasing width and depth improved some HORIZON scores but transferred inconsistently. We therefore replicated the stronger A/B models over three seeds and combined their predictions with fixed 50/50 weights (Table II). Every listed ensemble beat both components in all 24 matched seed/month

TABLE II
Multi-seed replication and fixed 50/50 ensembles

Model	HORIZON	HARVEST$_m$	HARVEST$_{30}$
A0b	4.267±0.018	4.497±0.024	4.491±0.024
A0l	4.236±0.014	4.488±0.026	4.581±0.048
B2b	4.289±0.004	4.430±0.038	4.414±0.093
B1b	4.283±0.005	4.472±0.096	4.471±0.056
A0b+B2b	4.144±0.009	4.378±0.028	4.355±0.068
A0l+B2b	4.131±0.006	4.380±0.017	4.406±0.057
A0b+B1b	4.142±0.011	4.396±0.038	4.390±0.017
A0l+B1b	4.130±0.004	4.406±0.067	4.436±0.042

Mean ± sample SD over seeds 17, 42, 123; b/l = base/large. HORIZON: eight-fold mean, best checkpoint, cap 50 and patience 10. HARVEST: median-best-epoch budget (m) or final epoch 30. All scores are MAE in EUR/MWh.

HORIZON comparisons. This supported combining different reconstruction choices rather than relying on a larger single model. Gated A0 and raw B2 predictions are averaged only after matching all date/zone/quarter keys.

V. Historical Context and Similar-Hour Retrieval

We next encoded yesterday's observed correction relative to MAKIMA,

$$c_{D-1,h,q} = y_{D-1,h,q} - H_{D-1,h} - m_{D-1,h,q},$$

and compared linear, CNN and Transformer history encoders. Table III shows that simple encoders were competitive: CNN led A0, while a history Transformer did not justify its additional complexity. History helped A-family models more consistently than B-family models. The initial fusion study changed the original attention blocks; we subsequently restored RoPE and used fresh matched controls.

The restored-backbone integration study tested additive and concatenative A0 injection, B2 FiLM and hourly correction

TABLE III
HISTORICAL D−1 ENCODER COMPARISON: HORIZON MAE

Base	None	LIN	CNN	Transformer
A0	4.429	4.358	4.315	4.356
A1	4.419	4.351	4.336	4.349
B1	4.294	4.344	4.328	4.342
B2	4.325	4.320	4.334	4.327

Seed 42; eight months; historical early-stopped Fourier-only fusion backbone. These controls differ from the original RoPE models. A-family scores include the training-fit gate.

TABLE IV
RETRIEVAL POLICIES WITH LINEAR/CNN ENCODING

Policy	A0 H	A0 T	B2 H	B2 T
R0-L	4.249	4.456	4.249	4.465
R0-C	4.253	4.482	4.253	4.571
R1-L	4.182	4.386	4.192	4.434
R1-C	4.175	4.414	4.180	4.410
R2-L	4.198	4.443	4.210	4.425
R2-C	4.175	4.444	4.201	4.460
R3-L	4.158	4.369	4.173	4.386
R3-C	4.146	4.331	4.187	4.439
R4-L	4.186	4.345	4.201	4.412
R4-C	4.173	4.390	4.212	4.481

H = eight-month HORIZON; T = HARVEST. L/C = linear/CNN. Seed 42; 30 epochs; H selects best validation checkpoint, T uses epoch 30. MAE in EUR/MWh; lower is better.

gates. Its best ensemble used *contextual A0 with context-free B2*; adding history to both components was not necessarily better. We then changed which history was retrieved. R0 uses yesterday; R1 chooses five whole days by standardized shape plus level/volatility regime; R2 compares whole-day gradients. R3 chooses five same-zone, same-clock-hour observations within 28 calendar days; R4 extends that local lookback to 90 days.

For query hour h and historical date j, R3/R4 use

$$\delta_{j,h}^2 = \frac{(\Delta_h^- - \Delta_{j,h}^-)^2 + (\Delta_h^+ - \Delta_{j,h}^+)^2}{2 \cdot 25^2} + 0.2\frac{(H_h - H_{j,h})^2}{60^2}.$$

The slope scales and price scales are in EUR/MWh. At day boundaries, the available one-sided slope is used twice. For the five nearest candidates, weights are $w_j \propto 1/\max(\delta_{j,h}, 10^{-8})$, normalized to sum to one; distances at most 10^{-12} share all weight. We use all available candidates if fewer than five exist. The aggregated corrections are recentered within each hour, encoded, and passed through a confidence gate based on distance, disagreement and neighbor count. R3 led HORIZON in both families, with CNN encoding for A0 and linear encoding for B2 (Table IV).

All candidates are strictly earlier same-zone dates. Training retrieves from training dates only; rolling evaluation may consume released observations from earlier dates in its own partition. Missing/DST dates are not synthesized. October 2025 has no supplied September history, so its first day lacks context and a 90-day lookback cannot provide 90 days of observations.

VI. FINAL CHOICE AND REFIT

The full grid (Fig. 2) reinforced the distinction between chronological validation and October transfer. HORIZON favored local A0 plus global B2 retrieval; HARVEST favored **A0-R3-CNN + B2-Original**. We selected the latter: **4.181 EUR/MWh HARVEST MAE**, a **19.31% improvement over MAKIMA**. The final weights remain 50/50. Its HORIZON macro/pooled-date MAEs are 4.064/4.072; the final choice explicitly uses the inspected transfer results.

Cross-attention over individual neighbors improved HORIZON in every matched seed/month comparison but worsened HARVEST at all three seeds; we discarded it from the final system. Longer paper-inspired SRT experiments also did not match the final ensemble's measured performance and remain separate exploratory work.

We refitted both final components from scratch on all 363 dates for exactly 30 epochs, seed 42, and froze the final checkpoints, normalization, A0 gate and weights. The full-data training score is not a new HARVEST estimate. Inference uses only the hourly curve and strictly earlier supplied history, with no retraining. Serialized outputs are checked for complete keys, finite values and hourly consistency. Neural runs retain resolved configurations, source/data identities and local checkpoints, with online Lightning/W&B metrics logging; classical runs remain local without W&B. Exact fold scores, pooled-date results and source hashes accompany this draft in the evidence notes.

TABLE V
FINAL HARVEST COMPARISON

Model	HARVEST MAE (EUR/MWh)
Flat baseline	8.944
MAKIMA baseline	5.182
A0-R3-CNN	4.331
B2-Original	4.318
A0-R3-CNN + B2-Original 50/50 ensemble	**4.181**